

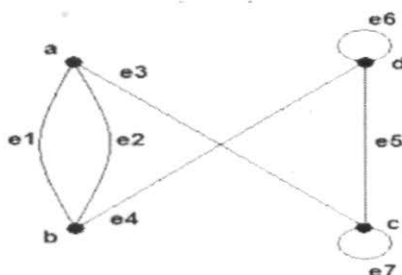
10. The Newton-Raphson formula to find the $(n+1)^{th}$ approximation to the root of $f(x) = 0$ is :
- A) $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ B) $x_{n+1} = x_n + \frac{f(x_n)}{f'(x_n)}$
 C) $x_{n+1} = x_n - \frac{f'(x_n)}{f(x_n)}$ D) $x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$
11. The method which approximates the curve of the tangent to find root of an algebraic equation is
- A) Regula Falsi Method B) Newton-Raphson Method
 C) Euler's Method D) Taylor's Method
12. For the heat equation $u_t = 0.5u_{xx}$, $h = k = 1$, the value of mesh ratio parameter α is
- A) 1 B) 0.5
 C) 0.25 D) 2
13. The Fourier series expansion of x^3 in the interval $-1 \leq x \leq 1$, periodic with $f(x+2) = f(x)$ has:
- A) Only Sine terms B) Only Cosine terms
 C) Both Sine and Cosine terms D) Only Sine terms and a non-zero constant
14. The Fourier series expansion of the function $f(x)$ in the interval $(\alpha, \alpha + 2L)$ is
- A) $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$ B) $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$
 C) $\frac{a_0}{2} + \sum_{n=0}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=0}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$ D) $a_0 + \sum_{n=0}^{\infty} a_n \cos(nx) + \sum_{n=0}^{\infty} b_n \sin(nx)$
15. Which of the following functions is odd
- A) $-x - x^3$ B) $\sin x^2 + x^5$
 C) $\cos 5x + 3e^{-x}$ D) $e^{-x} + e^x$
16. Fourier transform of the function $f(t) = \dots\dots\dots$
- A) $\int_{-\infty}^{\infty} f(t)e^{-\omega t} dt$ B) $\int_0^{\infty} f(t)e^{-i\omega t} dt$
 C) $\int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$ D) $\int_0^{\infty} f(t)e^{-\omega t} dt$
17. $Z(a^{nt})$ where a, t are constants is:
- A) $\frac{z}{z + a^t}$ B) $\frac{z}{z - a^t}$
 C) $\frac{z}{z + a^{-t}}$ D) $\frac{z}{z - a^{-t}}$
18. If $Z(u_n) = U(z)$, then $Z(-u_n)$ is :
- A) $U(-z)$ B) $-U(z)$
 C) $U\left(-\frac{1}{z}\right)$ D) $U\left(\frac{1}{z}\right)$
19. If a, b, c are constants and u_n, v_n, w_n are any discrete functions, then $Z(au_n + bv_n - cw_n)$ is
- A) $a Z(u_n) \frac{bZ(v_n)}{c} Z(w_n)$ B) $a Z(u_n) + bZ(v_n) - cZ(w_n)$
 C) $a Z(u_n) - bZ(v_n) - c Z(w_n)$ D) $a Z(u_n) bZ(v_n) c Z(w_n)$
20. If $Z(u_n) = U(z)$ then $\lim_{z \rightarrow \infty} U(z)$ is :
- A) 0 B) ∞
 C) u_0 D) u_1

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

	Marks	BT*	CO*	PO*
1. a) Using rules of Inference, Show that the premises "If you send me an e-mail message, then I will finish writing the program," "If you do not send me an e-mail message, then I will go to sleep early," and "If I go to sleep early, then I will wake up feeling refreshed" lead to the conclusion "If I do not finish writing the program, then I will wake up feeling refreshed."	6	L*2	1	1
b) Use mathematical induction to show $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for all non-negative integers n .	6	L2	1	2
c) Let p and q be the propositions "The election is decided" and "The votes have been counted," respectively. Express each of these compound propositions as an English sentence. i) $\neg p$ ii) $p \vee q$ iii) $\neg p \wedge q$ iv) $q \rightarrow p$	4	L3	1	1
2. a) Define 'tautology' and 'contingency'. By constructing a truth table verify the statement $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is a tautology or not?	6	L1	1	1
b) Prove that $\sqrt{2}$ is irrational by giving proof by contradiction.	6	L2	1	2
c) Show that the propositions $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalence.	4	L2	1	1

Unit – II

3. a) Let $A = \{1, 2, 3, 4, 5, 6\}$, R be a relation on A defined by aRb if and only if b is a multiple of a . i) Find a relation R ii) Give the matrix representation of R iii) Draw the digraph of R iv) Find the in-degree and out-degree of every vertex of R v) Find the domain and range of R .	6	L1	2	1
b) Let $A = \{1, 2, 3, 4\}$ and let $R = \{(1, 2), (2, 3), (3, 4), (2, 1)\}$. Find the transitive closure of R using Warshall's algorithm.	6	L2	2	2
c) Write the adjacency matrix and incidence matrix of the following graph.				



4. a) Determine whether the following relation R on a set A is an equivalent relation or not? A = set of real numbers; aRb if and only if $a^2 + b^2 = 4$.	6	L1	2	1
b) State handshaking theorem. Prove that undirected graph has an even number of vertices of odd degree.	6	L2	2	1
c) Define odd and even permutations. i) determine whether the permutation $p = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 2 & 1 & 6 & 5 & 8 & 7 & 3 \end{pmatrix}$ is even or odd ii) Find p^2 .	4	L1, L2	2	2

Unit – III

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|-------|--|---|----|---|---|
| 5. a) | Using Modified Euler's method find an approximate value of y when $x = 0.2$, given that $\frac{dy}{dx} = x + y$, $y(0) = 1$ and $h = 0.2$. Carry out 3 iterations. | 6 | L1 | 3 | 2 |
| b) | Solve Laplace equation $u_{xx} + u_{yy} = 0$ for $0 < x < 1$, $0 < y < 1$; $u(x, 0) = u(0, y) = 0$; $u(x, 1) = 6x$; $0 < x \leq 1$, $u(1, y) = 3y$, $0 < y < 1$. Divide the region into 9 square meshes. | 6 | L3 | 3 | 1 |
| c) | Find the root of the equation $5x = e^x$ in $(0, 1)$ using Regula-Falsi method correct to four decimal places. Carry out 4 iterations. | 4 | L2 | 3 | 1 |
| 6. a) | Using Runge-Kutta method solve the initial value problem $10y' = x^2 + y^2$, $y(0) = 1$, find $y(0.1)$ taking $h = 0.1$. | 6 | L3 | 3 | 1 |
| b) | Find the value of $u(x, t)$ satisfying the equation $\frac{\partial u}{\partial t} = 4 \frac{\partial^2 u}{\partial x^2}$ and the boundary condition $u(0, t) = 0 = u(8, t)$; $u(x, 0) = 4x - \frac{1}{2}x^2$; $x = i$, $i = 0, 1, 2, 3, 4, 5, 6, 7, 8$, $t = \frac{1}{8}j$, $j = 0, 1, 2, 3, 4, 5$. | 6 | L1 | 3 | 1 |
| c) | Find an approximate value of y when $x = 0.1$, if $y' = x^2y - 1$, $y(0) = 1$ and $h = 0.1$. Using Taylor's series method upto third degree terms. | 4 | L1 | 3 | 1 |

Unit – IV

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|-------|---|---|--------|---|---|
| 7. a) | Find the Fourier Series expansion of $f(x) = \begin{cases} 0 & -2 < x < 0 \\ 1 & 0 < x < 2 \end{cases}$; $f(x+4) = f(x)$. | 6 | L3 | 4 | 1 |
| b) | Find the Fourier series for $f(x) = e^{-x}$ in the interval $0 < x < 2\pi$. | 6 | L2 | 4 | 1 |
| c) | Find the half range Fourier cosine series of $f(x) = x^2$ in $0 \leq x \leq \pi$. | 4 | L1 | 4 | 1 |
| 8. a) | Find the Fourier transforms of $f(x) = e^{-a x }$. | 6 | L1 | 4 | 1 |
| b) | State and prove convolution theorem for Fourier transforms. | 6 | L1, L3 | 4 | 2 |
| c) | Find the Fourier sine transform of $\frac{e^{-ax}}{x}$, $a > 0$. | 4 | L1 | 4 | 1 |

Unit – V

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|--------|--|---|--------|---|---|
| 9. a) | Find the Z-transform of
i) $\cos n\theta$ ii) $\sin n\theta$ | 6 | L1 | 5 | 1 |
| b) | State and prove convolution theorem for Z-transform. | 6 | L1, L3 | 5 | 2 |
| c) | Find Z-transform of $(2^n + 2^n n + 3^{-n} n + 3)$. | 4 | L1 | 5 | 1 |
| 10. a) | Find the inverse Z-transform of $\frac{z^2}{(z^2 - 7z + 12)}$. | 6 | L2 | 5 | 1 |
| b) | Using Z-transform solve the difference equation $u_{n+2} - 4u_n = 0$, given $u_0 = 0$ and $u_1 = 2$. | 6 | L2 | 5 | 1 |
| c) | Prove that $Z_T(n^k) = -z \frac{d}{dz} (Z_T(n^{k-1}))$, where k is a positive integer. | 4 | L1 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
